

Bring on the buttons!

In the next stage you'll make the buttons and add them to the GUI. Then you'll create a function called `show_symbol()` to control what happens when a player clicks on the buttons.

8 Build the grid

The grid will consist of 24 buttons arranged into four rows of six. To lay out the grid, you'll use nested loops. The outer `x` loop will work from left to right across the six columns, while the inner `y` loop will work from top to bottom down each column. Once the loops have run, each button will have been given a pair of `x` and `y` coordinates that set its position on the grid. Put this block of code after the `shuffle` command.

```
random.shuffle(symbols)
```

```
for x in range(6):
```

```
    for y in range(4):
```

```
        button = Button(command=lambda x=x, y=y: show_symbol(x, y), \
                        width=3, height=3)
```

```
        button.grid(column=x, row=y)
```

```
        buttons[x, y] = button
```

```
        button_symbols[x, y] = symbols.pop()
```

These are nested loops.

This line creates each button and sets its size and action when pressed.

The button is placed on the GUI.

Use a backslash character if you need to split a long line of code over two lines.

This line saves each button in the `buttons` dictionary.

The button's symbol is set by this line.

△ How it works

Each time the loop runs, the `lambda` function saves the current button's `x` and `y` values (the row and column it's in). When the button is pressed, it calls the `show_symbol()` function (which you'll create later) with these values, so the function knows which button has been pressed and which symbol to reveal.



EXPERT TIPS

Button

Tkinter has a built-in widget called **Button**, which we use to create the GUI buttons. You can pass different parameters to it. The ones we need are **command**, **width**, and **height**. The **command** parameter tells the program what to do when a button is pressed. This is a function call. In our program, it calls a **lambda** function. The **width** and **height** parameters are used to set the size of the button.

